

Covid-19 Classification

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By Landon Nguyen

Background Information

Research Question:

- Can deep learning be used on chest x-rays to differentiate between healthy individuals and those infected with Covid-19 or viral pneumonia?

Goal:

- Use chest x-rays to differentiate between Covid-19, healthy, and viral pneumonia.

Purpose:

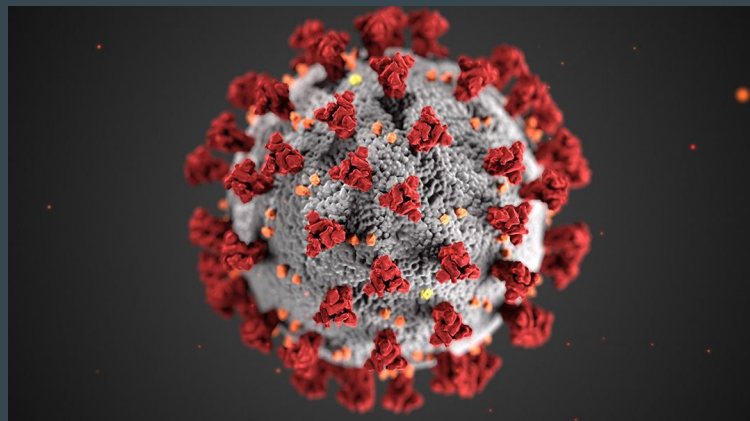
- X-rays can provide an efficient way of diagnosing numerous people together.
- Standard viral nucleic acid detection is sensitive to disease progression, immune status, etc.

Dataset:

<https://www.kaggle.com/datasets/pranavraikokte/covid19-image-dataset>

Setting a Precedent for AI to:

- Diagnose other diseases (ie. cancer)
- Classify case severity
- Introduce AI into medicinal practices



From: <https://www.cdc.gov/museum/timeline/covid19.html>

Exploration

EDA/Challenges/Standard Setup

Data Size and Split:

- Total: 317
 - Very small
- Train: 251 images (~80%)
- Validation: 33 images (~10%)
- Test: 33 images (~10%)

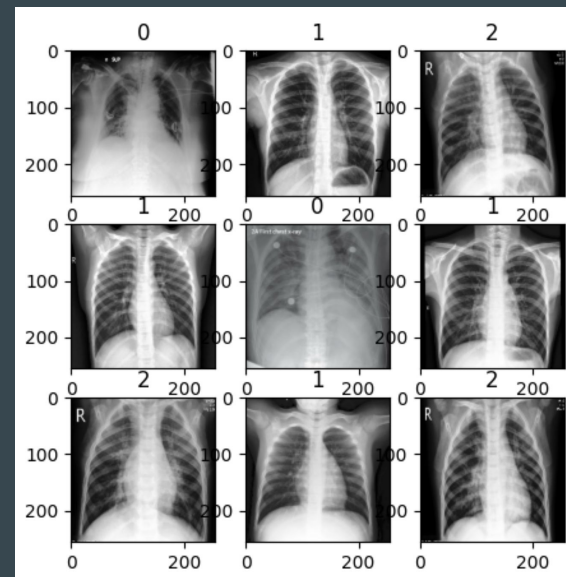
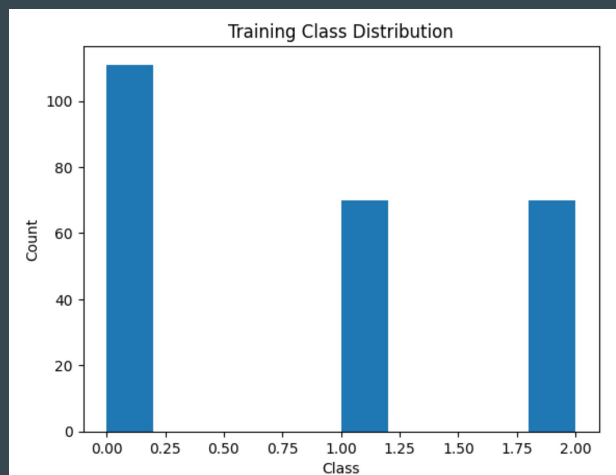
Train Class Distribution:

- Covid-19 (0): 111 images ~44%
- Normal (1): 70 images ~28%
- Pneumonia (2): 70 images ~28%
- Slight Imbalance

* Images were rescaled between 0 and 1

Target Metrics:

- Macro F1:
 - Replaces accuracy on an imbalanced dataset
- Recall:
 - Diagnosing a highly contagious disease
 - Goal is to minimize false negatives



ANN Baseline

Design:

- Flatten
- One hidden layer
- Output

Optimization/Fitting:

- Adam (LR 0.001)
- Epochs: 20
- Batch Size: 32

Model: "sequential"

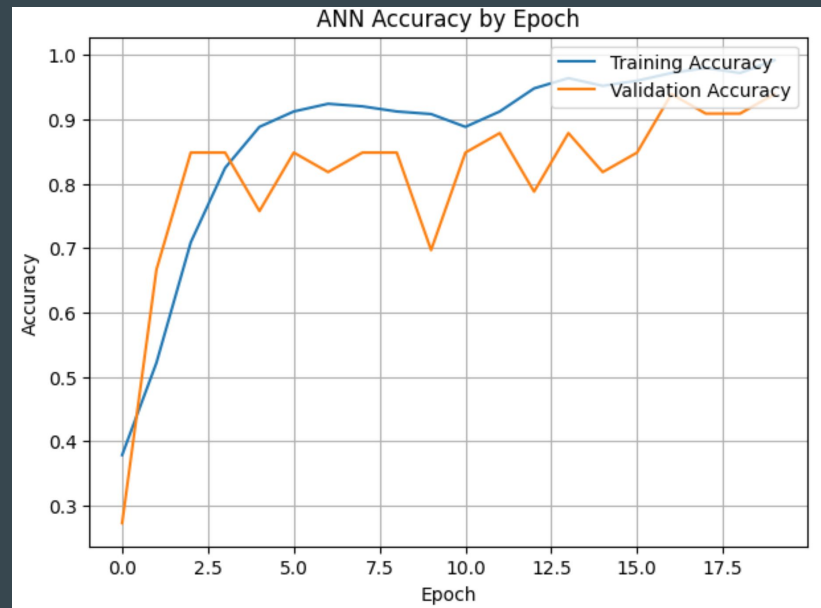
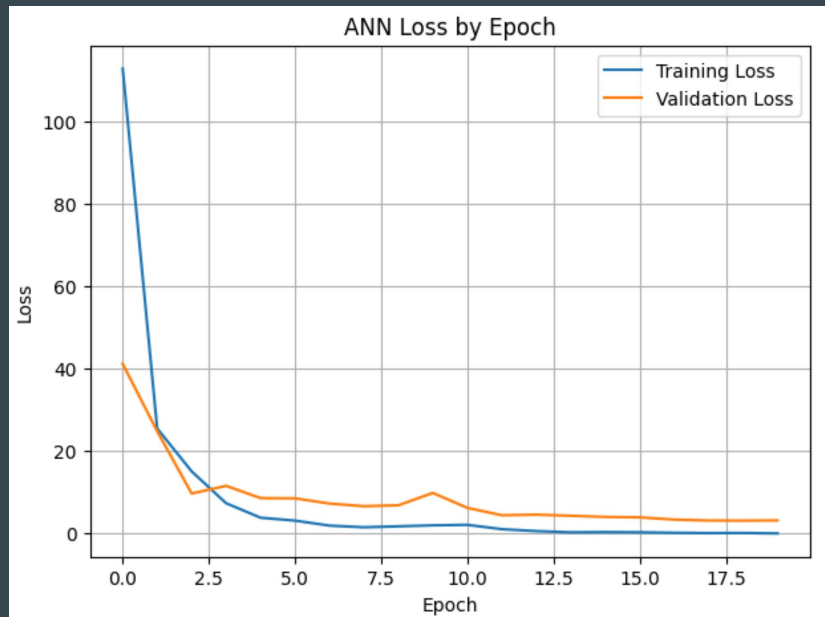
Layer (type)	Output Shape	Param #
flatten (Flatten)	(None , 196608)	0
dense (Dense)	(None , 128)	25,165,952
dense_1 (Dense)	(None , 3)	387

Total params: 25,166,339 (96.00 MB)

Trainable params: 25,166,339 (96.00 MB)

Non-trainable params: 0 (0.00 B)

ANN Baseline Results



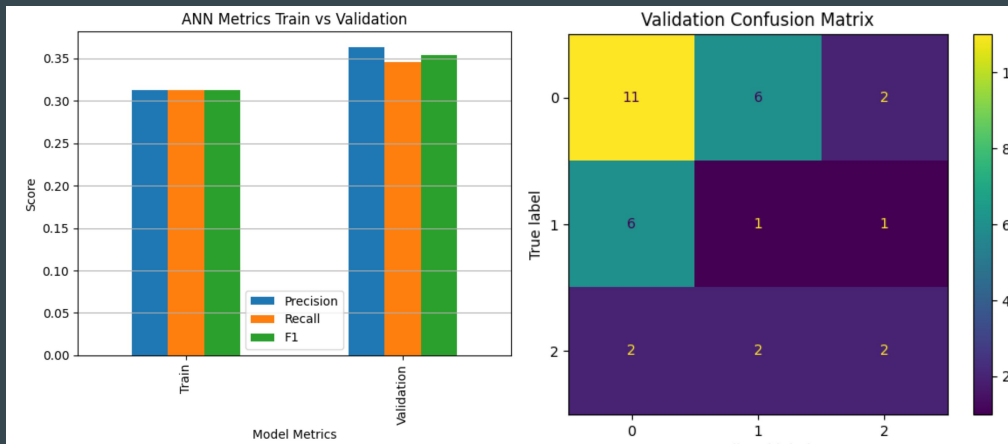
ANN Baseline Results

Train Report:

	precision	recall	f1-score	support
0	0.43	0.42	0.43	111
1	0.27	0.29	0.28	70
2	0.24	0.23	0.23	70
accuracy			0.33	251
macro avg	0.31	0.31	0.31	251
weighted avg	0.33	0.33	0.33	251

Validation Report:

	precision	recall	f1-score	support
0	0.58	0.58	0.58	19
1	0.11	0.12	0.12	8
2	0.40	0.33	0.36	6
accuracy			0.42	33
macro avg	0.36	0.35	0.35	33
weighted avg	0.43	0.42	0.43	33



CNN Baseline

Design:

- One layer of convolution and pooling
- Flattening
- One hidden layer
- Output

Optimization/Fitting:

- Adam (LR 0.001)
- Epochs: 20
- Batch Size: 32

Model: "sequential_2"

Layer (type)	Output Shape	Param #
conv2d_1 (Conv2D)	(None, 256, 256, 16)	448
max_pooling2d_1 (MaxPooling2D)	(None, 128, 128, 16)	0
flatten_2 (Flatten)	(None, 262144)	0
dense_4 (Dense)	(None, 128)	33,554,560
dense_5 (Dense)	(None, 3)	387

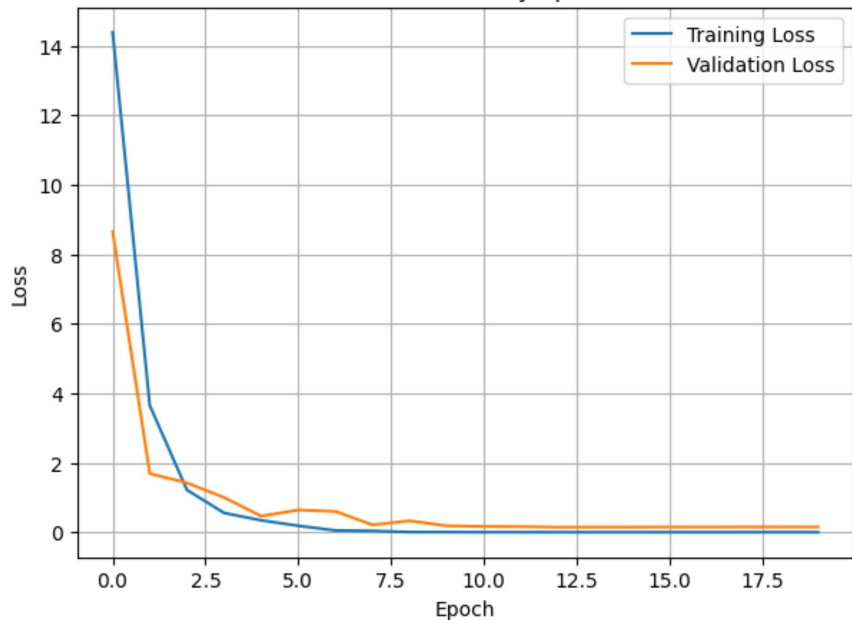
Total params: 33,555,395 (128.00 MB)

Trainable params: 33,555,395 (128.00 MB)

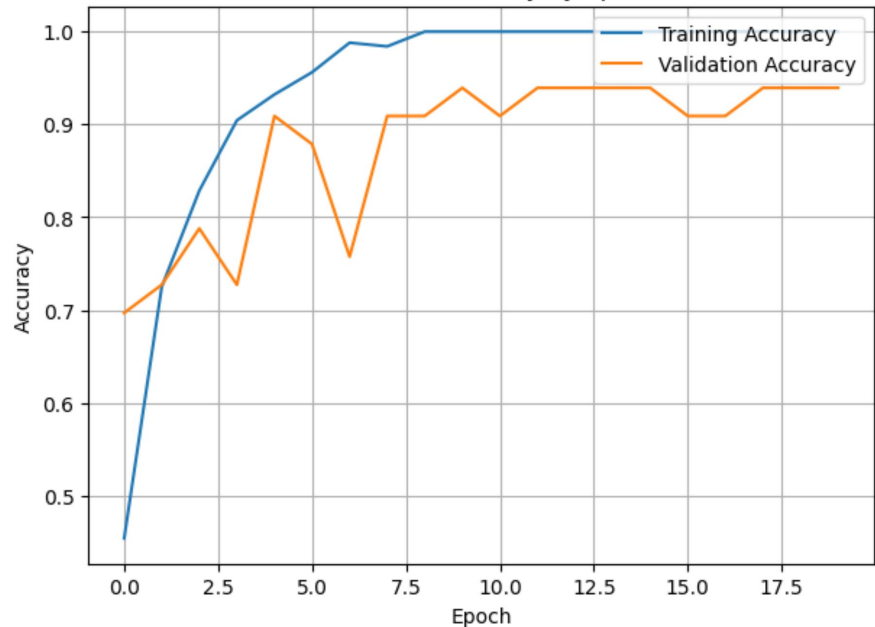
Non-trainable params: 0 (0.00 B)

CNN Baseline Results

Base CNN Loss by Epoch



Base CNN Accuracy by Epoch



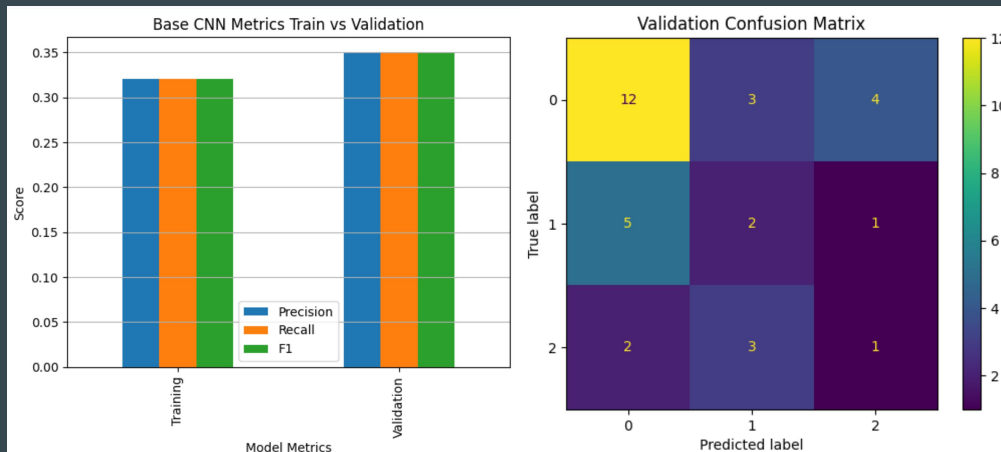
CNN Baseline Results

Training Report:

	precision	recall	f1-score	support
0	0.43	0.43	0.43	111
1	0.31	0.31	0.31	70
2	0.21	0.21	0.21	70
accuracy			0.34	251
macro avg	0.32	0.32	0.32	251
weighted avg	0.34	0.34	0.34	251

Validation Report:

	precision	recall	f1-score	support
0	0.63	0.63	0.63	19
1	0.25	0.25	0.25	8
2	0.17	0.17	0.17	6
accuracy			0.45	33
macro avg	0.35	0.35	0.35	33
weighted avg	0.45	0.45	0.45	33



Preprocessing

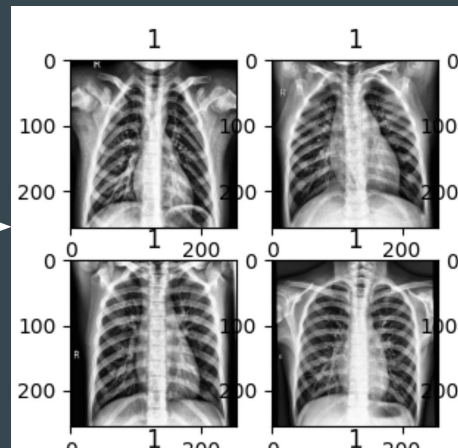
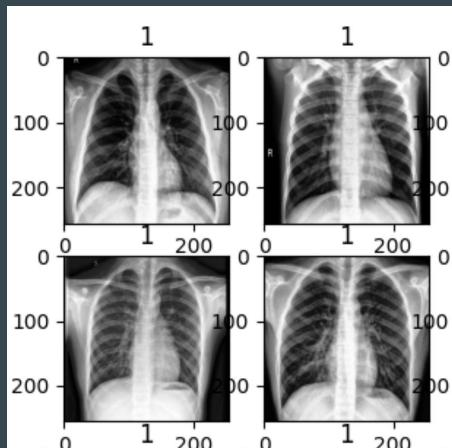
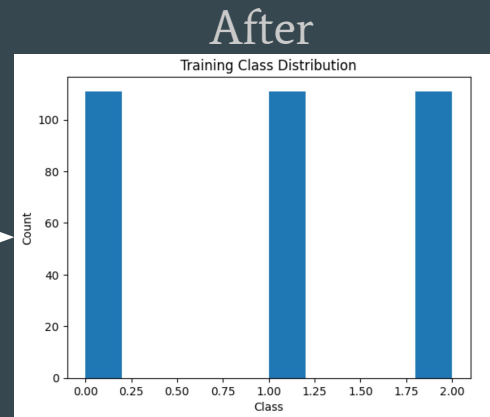
Image Augmentation and Imbalance Handling

The goal:

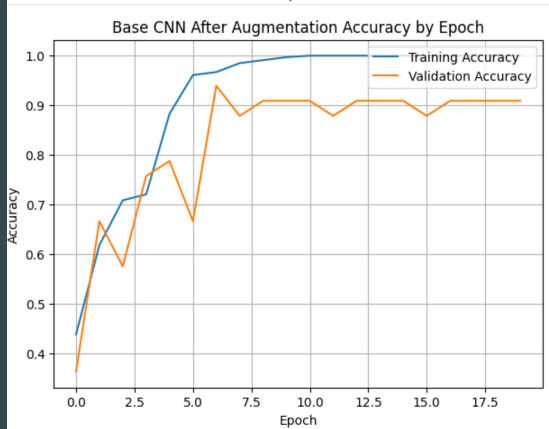
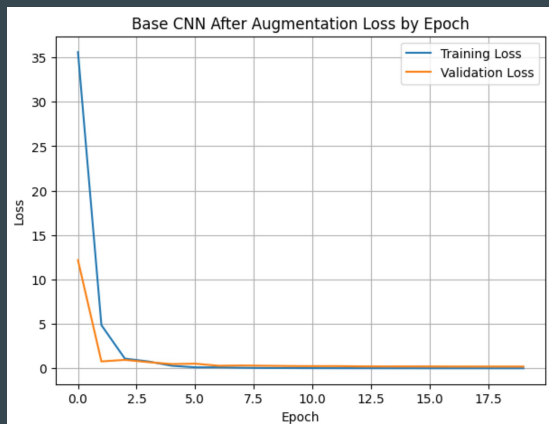
- To mimic simple random oversampling with the addition of image augmentation.

Steps taken:

1. Take a small random subset of a minority class (normal or pneumonia)
2. Apply CLAHE (Contrast Limited Adaptive Histogram Equalization)
3. Add to the dataset



Baseline Update

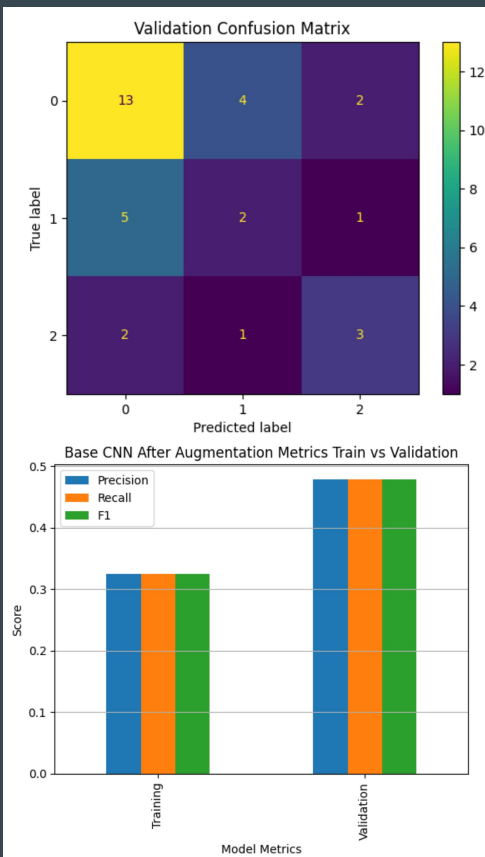


Training Report:

	precision	recall	f1-score	support
0	0.43	0.43	0.43	111
1	0.26	0.26	0.26	70
2	0.29	0.29	0.29	70
accuracy			0.34	251
macro avg	0.33	0.33	0.33	251
weighted avg	0.34	0.34	0.34	251

Validation Report:

	precision	recall	f1-score	support
0	0.65	0.68	0.67	19
1	0.29	0.25	0.27	8
2	0.50	0.50	0.50	6
accuracy			0.55	33
macro avg	0.48	0.48	0.48	33
weighted avg	0.53	0.55	0.54	33



Advanced Modeling

Advanced CNN Model

Design:

- Two layers of convolution and pooling
- Flatten
- Two hidden layers
- Dropout: 0.5
- Output

Optimization/Fitting:

- Adam (LR 0.001)
- Epochs: 20
- Batch Size: 32

Layer (type)	Output Shape	Param #
conv2d_3 (Conv2D)	(None, 256, 256, 16)	448
max_pooling2d_3 (MaxPooling2D)	(None, 128, 128, 16)	0
conv2d_4 (Conv2D)	(None, 128, 128, 32)	4,640
max_pooling2d_4 (MaxPooling2D)	(None, 64, 64, 32)	0
flatten_4 (Flatten)	(None, 131072)	0
dense_8 (Dense)	(None, 128)	16,777,344
dropout (Dropout)	(None, 128)	0
dense_9 (Dense)	(None, 64)	8,256
dropout_1 (Dropout)	(None, 64)	0
dense_10 (Dense)	(None, 3)	195

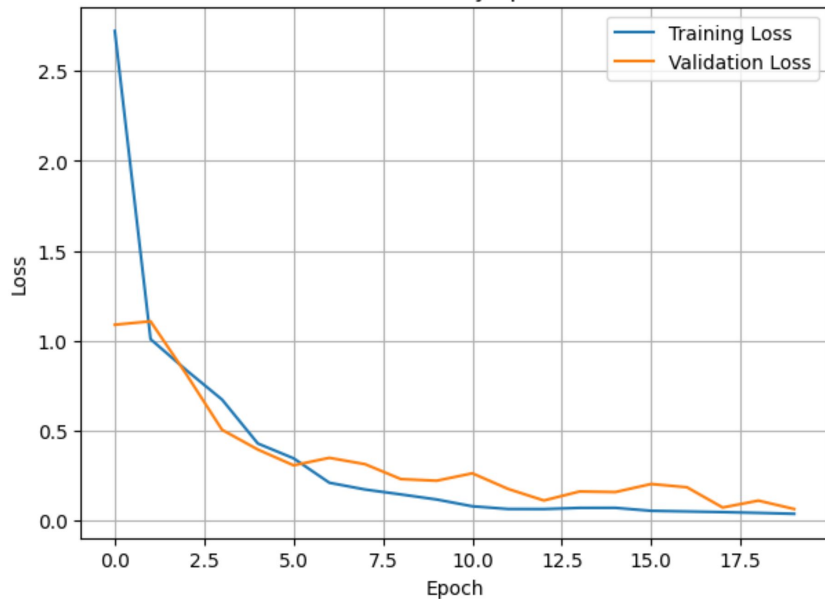
Total params: 16,790,883 (64.05 MB)

Trainable params: 16,790,883 (64.05 MB)

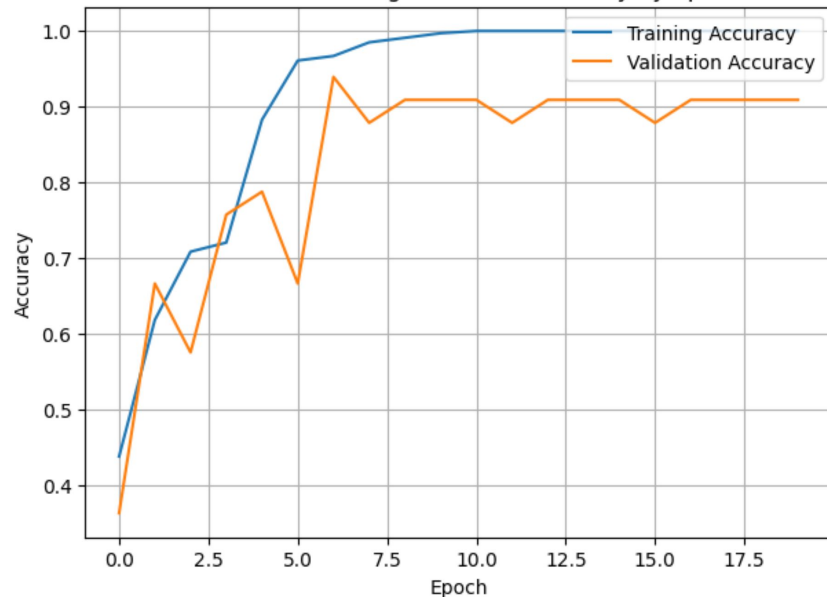
Non-trainable params: 0 (0.00 B)

Advanced CNN Results

CNN Loss by Epoch



Base CNN After Augmentation Accuracy by Epoch



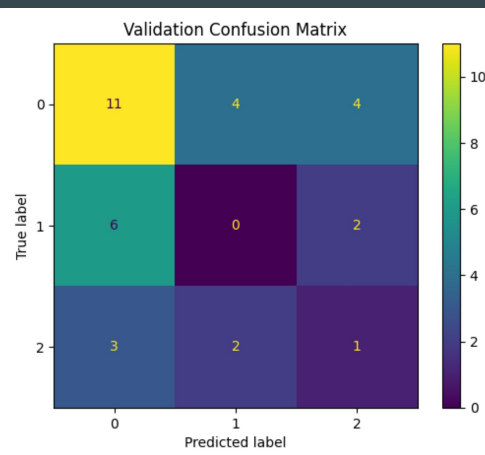
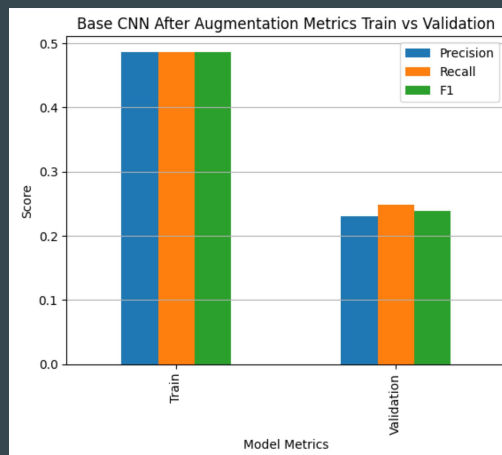
Advanced CNN Results

Train Report:

	precision	recall	f1-score	support
0	0.40	0.40	0.40	111
1	0.57	0.57	0.57	111
2	0.50	0.50	0.50	111
accuracy			0.49	333
macro avg	0.49	0.49	0.49	333
weighted avg	0.49	0.49	0.49	333

Validation Report:

	precision	recall	f1-score	support
0	0.55	0.58	0.56	19
1	0.00	0.00	0.00	8
2	0.14	0.17	0.15	6
accuracy			0.36	33
macro avg	0.23	0.25	0.24	33
weighted avg	0.34	0.36	0.35	33



Transfer Learning

VGG16

Design:

- VGG16 (Frozen)
- Flatten
- One hidden layer
- Output

Optimization and Training

- Adam (LR 0.001)
- Epochs: 10
- Batch Size: 32

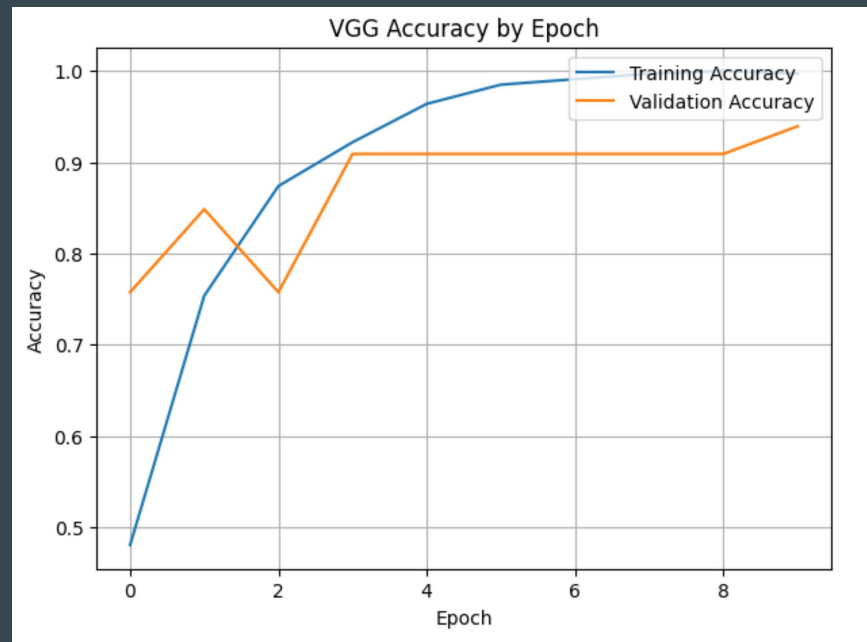
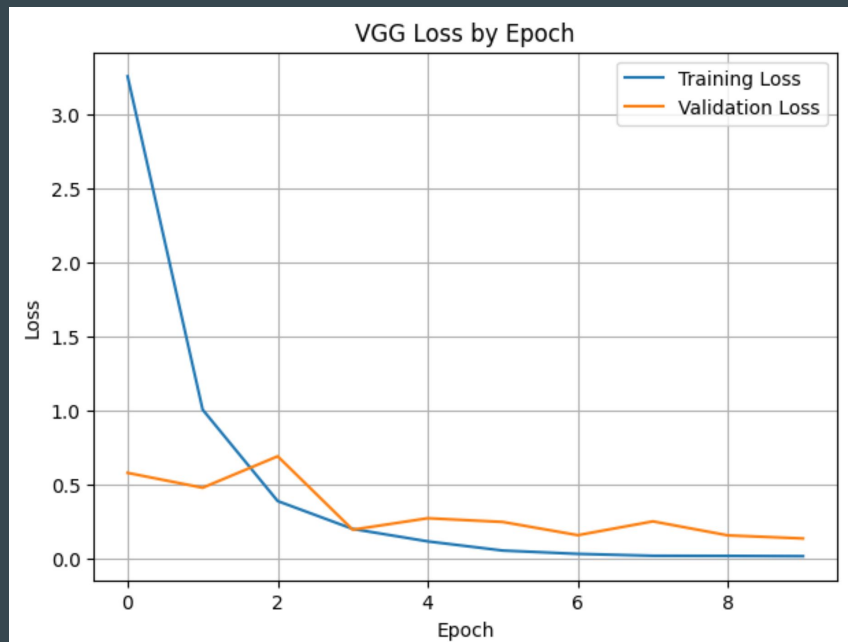
Layer (type)	Output Shape	Param #
vgg16 (Functional)	(None , 8, 8, 512)	14,714,688
flatten_2 (Flatten)	(None , 32768)	0
dense_4 (Dense)	(None , 128)	4,194,432
dense_5 (Dense)	(None , 3)	387

Total params: 18,909,507 (72.13 MB)

Trainable params: 4,194,819 (16.00 MB)

Non-trainable params: 14,714,688 (56.13 MB)

VGG16 Results



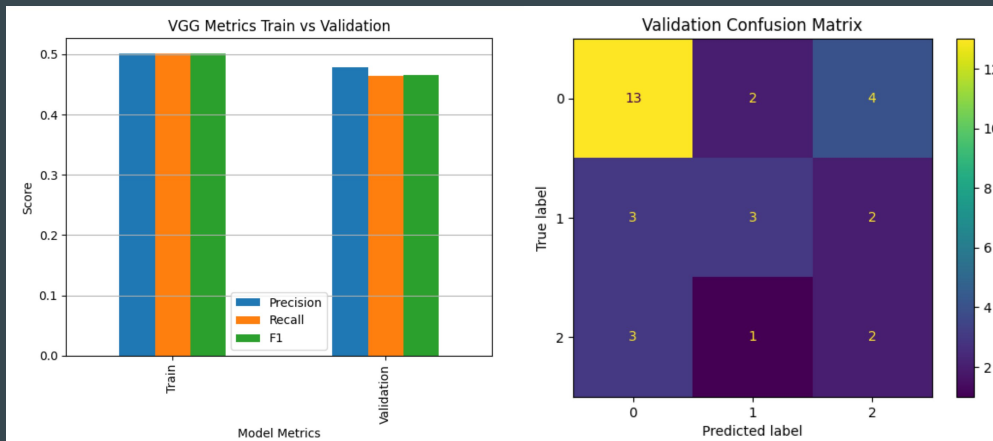
VGG16 Results

Train Report:

	precision	recall	f1-score	support
0	0.46	0.46	0.46	111
1	0.52	0.52	0.52	111
2	0.52	0.52	0.52	111
accuracy			0.50	333
macro avg	0.50	0.50	0.50	333
weighted avg	0.50	0.50	0.50	333

Validation Report:

	precision	recall	f1-score	support
0	0.68	0.68	0.68	19
1	0.50	0.38	0.43	8
2	0.25	0.33	0.29	6
accuracy			0.55	33
macro avg	0.48	0.46	0.47	33
weighted avg	0.56	0.55	0.55	33



DenseNet201

Augmentation:

- Random horizontal flipping
- Shear (0.1)
- Zoom (0.1)
- Rotation (15)
- Width/Height shifting (0.1)

Data Split:

- Train: 201 images ~ 63%
- Validation: 50 images ~ 16%
- Test: 66 images ~ 21%

Layer (type)	Output Shape	Param #	Connected to
input_layer_4 (InputLayer)	(None, 224, 224, 3)	0	-
zero_padding2d_6 (ZeroPadding2D)	(None, 230, 230, 3)	0	input_layer_4[0]...
conv1_conv (Conv2D)	(None, 112, 112, 64)	9,408	zero_padding2d_6...
conv1_bn (BatchNormalizatio...	(None, 112, 112, 64)	256	conv1_conv[0][0]
conv1_relu (Activation)	(None, 112, 112, 64)	0	conv1_bn[0][0]

...

bn (BatchNormalizatio...	(None, 7, 7, 1920)	7,680	conv5_block32_co...
relu (Activation)	(None, 7, 7, 1920)	0	bn[0][0]
global_average_poo... (GlobalAveragePool...	(None, 1920)	0	relu[0][0]
dropout_2 (Dropout)	(None, 1920)	0	global_average_p...
dense_2 (Dense)	(None, 3)	5,763	dropout_2[0][0]

Total params: 18,327,747 (69.91 MB)

Trainable params: 5,763 (22.51 KB)

Non-trainable params: 18,321,984 (69.89 MB)

DenseNet201

Design:

- DenseNet201
 - 43 trainable layers
- Global Average Pooling
- Dropout: 0.3
- Output

Optimization/Training:

- Adam (LR 0.00001)
- Epochs: 25
- Batch Size: 16
- Class Weights:

```
{0: np.float64(0.752)
1: np.float64(1.196)
2: np.float64(1.196)
```

Strategy:

- Unfreeze layers
 - Allow the model to learn more about the covid dataset

Targeted Challenges:

- Dataset is too small to create an effective model from scratch
- ImageNet is not optimized for x-ray images

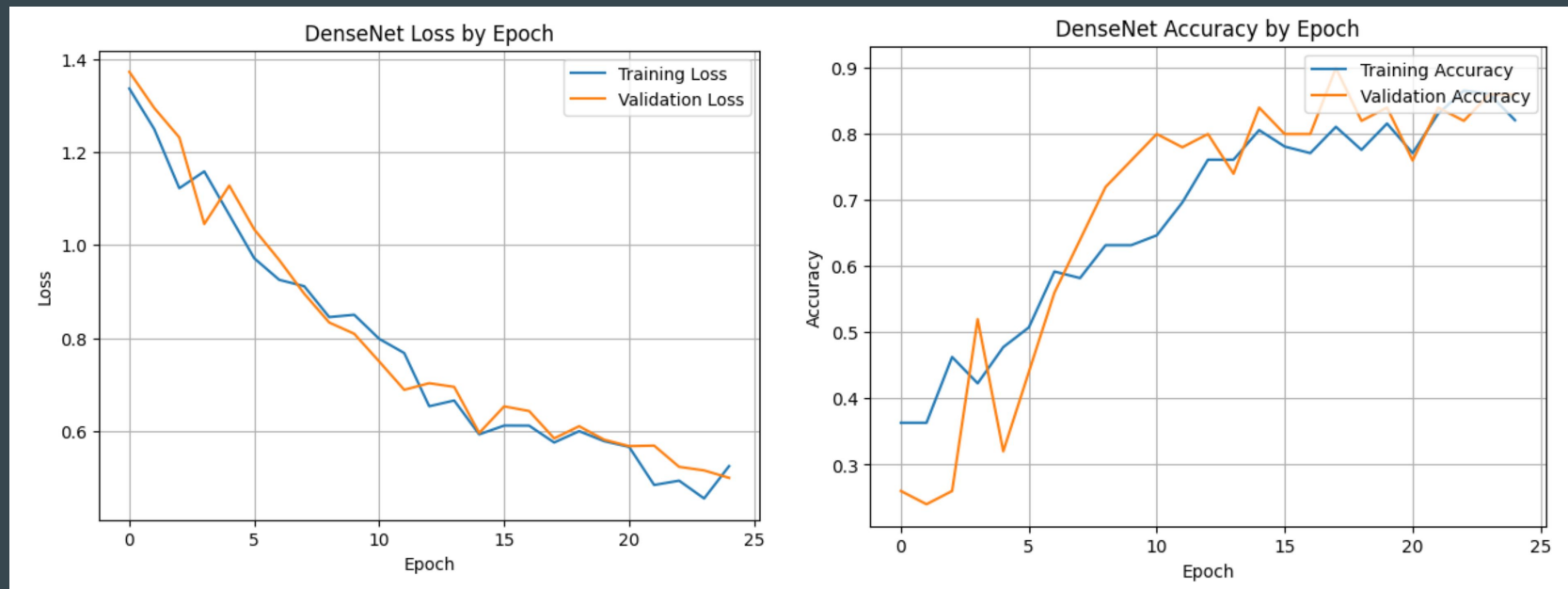
Layer (type)	Output Shape	Param #	Connected to
input_layer_4 (InputLayer)	(None, 224, 224, 3)	0	-
zero_padding2d_6 (ZeroPadding2D)	(None, 230, 230, 3)	0	input_layer_4[0]...
conv1_conv (Conv2D)	(None, 112, 112, 64)	9,408	zero_padding2d_6...
conv1_bn (BatchNormalizatio...	(None, 112, 112, 64)	256	conv1_conv[0][0]
conv1_relu (Activation)	(None, 112, 112, 64)	0	conv1_bn[0][0]

...

bn (BatchNormalizatio...	(None, 7, 7, 1920)	7,680	conv5_block32_co...
relu (Activation)	(None, 7, 7, 1920)	0	bn[0][0]
global_average_poo... (GlobalAveragePool...	(None, 1920)	0	relu[0][0]
dropout_2 (Dropout)	(None, 1920)	0	global_average_p...
dense_2 (Dense)	(None, 3)	5,763	dropout_2[0][0]

Total params: 18,327,747 (69.91 MB)
Trainable params: 5,763 (22.51 KB)
Non-trainable params: 18,321,984 (69.89 MB)

DenseNet201 Results



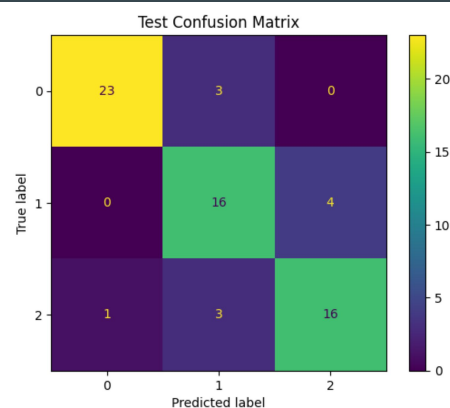
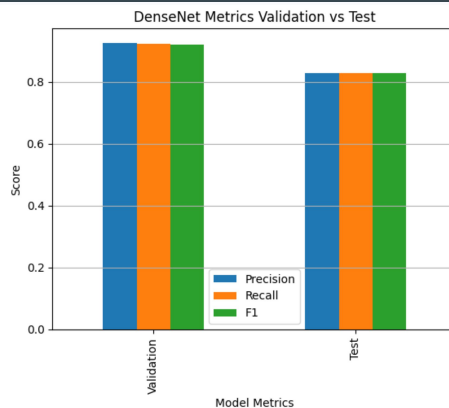
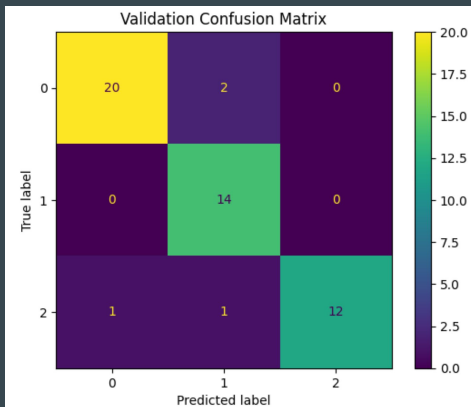
DenseNet201 Results

Validation Report:

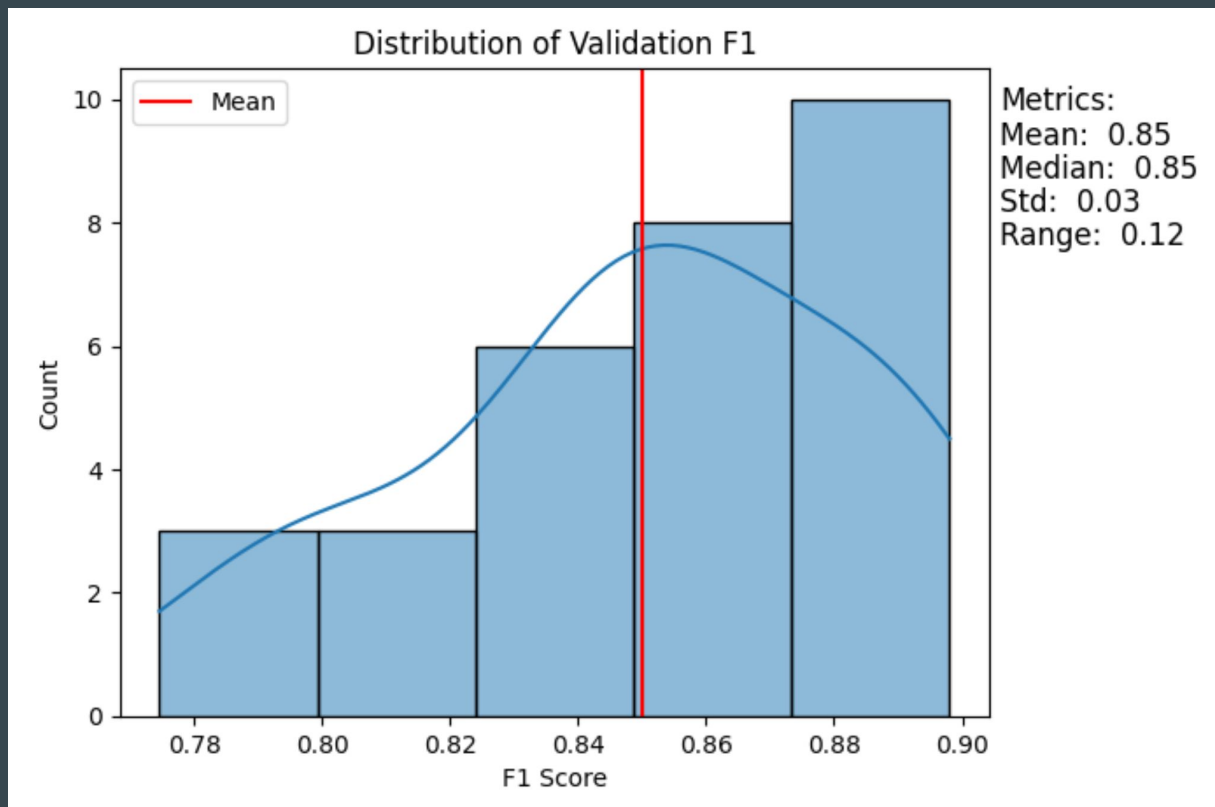
	precision	recall	f1-score	support
0	0.95	0.91	0.93	22
1	0.82	1.00	0.90	14
2	1.00	0.86	0.92	14
accuracy			0.92	50
macro avg	0.93	0.92	0.92	50
weighted avg	0.93	0.92	0.92	50

Test Report:

	precision	recall	f1-score	support
0	0.96	0.88	0.92	26
1	0.73	0.80	0.76	20
2	0.80	0.80	0.80	20
accuracy			0.83	66
macro avg	0.83	0.83	0.83	66
weighted avg	0.84	0.83	0.84	66

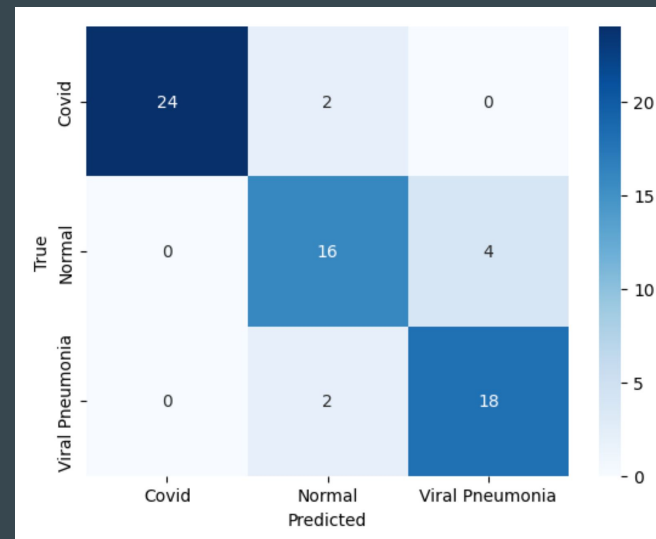


DenseNet201 Distribution of Validation F1



DenseNet201 Highest Test Results

	precision	recall	f1-score	support
Covid	1.00	0.92	0.96	26
Normal	0.80	0.80	0.80	20
Viral Pneumonia	0.82	0.90	0.86	20
accuracy			0.88	66
macro avg	0.87	0.87	0.87	66
weighted avg	0.88	0.88	0.88	66



Conclusion/Future Work

Are the models usable?

- High variance/inconsistent performance
- Perhaps for a probably positive/negative classification
- Good recall for both Covid-19 and viral pneumonia – low false negative rate

Model weaknesses:

- Neither ImageNet nor DenseNet are particularly optimal for x-ray images
- Slight imbalance between the classes

Future work:

- Attempting different augmentation methods
- Imbalance handling
- Using a more optimal transfer learning model

Future Goals:

- Pushing macro F1 past 90%
- Minimizing the range and standard deviation of validation F1

Technique(s)	Classification type	Precision	Sensitivity	Specificity	F1-score	Accuracy
Deep Bayes-SqueezeNet (COVIDiagnosis-Net)	Multi-class (COVID, Normal, Pneumonia)	N/A	N/A	99.10%	98.30%	98.26%

Kumar S, Shastri S, Mahajan S, et al. LiteCovidNet: A lightweight deep neural network model for detection of COVID-19 using X-ray images. *Int J Imaging Syst Technol.* 2022; 32(5): 1464-1480. doi:[10.1002/ima.22770](https://doi.org/10.1002/ima.22770)

Results from another similar study

Thank You!
Questions?